

REPSOL | DIELECTRIC OILS

Repsol Electra 3X Plus

Paraffinic (isoparaffinic) dielectric oil — Inhibited

PRODUCT DATA SHEET

Description

ELECTRA 3X PLUS is an oil formulated with isoparaffinic base stock, designed for use as insulating fluid in electrical equipment such as power and distribution transformers, circuit breakers, rheostats, etc. It is manufactured from highly-refined treated base oils ensuring the lack of any solid, polar or water compounds, and the hydrogenation process guarantees the absence of corrosive sulphur.

Classification and standards

ELECTRA 3X PLUS is an inhibited oil that complies with the IEC 60296 and ASTM D3487 Type II standards. It also complies with Section 7.1 of the IEC 60296 standard (Higher oxidation stability and low sulphur content).

Technical specifications

Values according to IEC 60296: 2012 Inhibited, including section 7.1 "Greater oxidative stability and low sulphur content".

Properties	Units	Method	Min.	Max.	Typical value
Physical properties					
Appearance		Visual		Bright and clear	
Density at 20 °C	g/cm3	ISO 12185		0.895	0.824
Viscosity at 100 °C	mm2/s	ASTM D445			3.1
Viscosity at 40 °C	mm2/s	ASTM D445		12	11.0
Viscosity at -30 °C	mm2/s	ASTM D445		1,800	379
Freezing point	°C	ISO 3016		-40	-67
Interfacial tension	mN/m	UNE 21320	40		54
Chemical properties					
Acidity	mg KOH/g	IEC 62021-2		0.01	<0.01
Corrosive sulphur		DIN 51353		Non-corrosive	
Corrosive sulphur		ASTM D1275B		Non-corrosive	
Potentially corrosive sulphur		IEC 62535		Non-corrosive	
Total sulphur content	%	ASTM D2622		0.050	<0.0001
Water content	mg/kg	IEC 60814		30	9.2
Inhibitor content	% weight	IEC 60666		0.4	0.3
DBDS	mg/kg	IEC 62697-1		Not detectable	
Furfural content	mg/kg	IEC 61198		Not detectable (<0.05)	
Electrical properties					

Properties	Units	Method	Min.	Max.	Typical value
Dielectric loss factor at 90 °C		IEC 60247		0.005	0.0003
Breakdown voltage, untreated	kV	IEC 60156	30		65
Breakdown voltage, treated	kV	IEC 60156	70		>70
Oxidation stability — IEC 61125C (500 h / 120 °C)					
Total acidity	mg KOH/g	IEC 61125C		0.3	0.14
Sludge	% weight	IEC 61125C		0.05	<0.01
Dielectric loss factor at 90 °C		IEC 61125C		0.05	0.0005
Health, safety and the environment					
Flash point	°C	ASTM D93	135		195
PCA content	% weight	IP 346		3	0.8
PCB content	% weight	IEC 61619	Exempt		Not detected

Note: Typical values correspond to current production. Whilst future production will conform to Repsol's specification, variations in these characteristics may occur. The guaranteed minimum/maximum values define the product specification.